Ka of Weak Acids by Titration

About 110 minutes

In this lab you find the acid dissociation constant, K_a, of two weak acids: acetic acid in distilled white vinegar and oxalic acid extracted from a powdered cleanser. K_a measures how far an acid dissociates in water and it sets the pH of every buffer, blood included. You will make a dilute sodium hydroxide solution from drain-cleaner lye, titrate each acid by weighing the titrant delivered from a dropper bottle, and then prepare mixtures that are one quarter, one half and three quarters of the way to the equivalence point. The pH of those mixtures, read from test strips, gives the pK_a directly.

> **Why the half-way point gives pKa (optional).** When half of a weak acid HA has been converted to its conjugate base A^-, the two are present in equal amounts, and the Henderson–Hasselbalch equation reduces to pH = pK_a. The concentration of the sodium hydroxide never enters this result. Only the *fraction* of the acid that has been neutralized matters, and that fraction is fixed by the mass of titrant added divided by the mass needed to reach the equivalence point. The method is insensitive to small errors in that mass, because the pH of a buffer changes slowly with composition. The result is only as good as the pH reading, so the accuracy of this lab is set by the test strips.

Equations

Dissociation of a monoprotic weak acid and its equilibrium constant:

$CH_3COOH(aq) \rightleftharpoons H^+(aq) + CH_3COO^-(aq)$

$$K_a = \frac{[\mathrm{H^+}][\mathrm{A^-}]}{[\mathrm{HA}]} \qquad \mathrm{p}K_a = -\log K_a$$

where $[H^+]$, $[A^-]$ and [HA] are equilibrium concentrations in mol/L.

Henderson–Hasselbalch equation:

$$\mathrm{pH} = \mathrm{p}K_a + \log \frac{[\mathrm{A^-}]}{[\mathrm{HA}]}$$

where pH is the measured pH of the mixture and $[A^-]/[HA]$ is the mole ratio of conjugate base to undissociated acid.

Fraction of the acid neutralized:

$$R = \frac{m_{eq}}{m_v} \qquad f = \frac{m_t}{R\, m_s}$$

where m_{eq} is the mass of titrant that reached the equivalence point in g, m_v is the mass of acid sample in that titration in g, R is the titrant-to-sample ratio in g/g, m_s is the mass of acid sample in the pH mixture in g, m_t is the mass of titrant added to the pH mixture in g, and f is the fraction of equivalence reached (dimensionless).

Base-to-acid ratio for a monoprotic acid (acetic acid, 0 < f < 1):

$$\frac{[\mathrm{A^-}]}{[\mathrm{HA}]} = \frac{f}{1-f}$$

Base-to-acid ratio for the second dissociation of a diprotic acid (oxalic acid, 0.5 < f < 1, where *f* is the fraction of the *second* equivalence point):

$HC_2O_4^-(aq) \rightleftharpoons H^+(aq) + C_2O_4^{2-}(aq)$

$$\frac{[C_2O_4^{2-}]}{[HC_2O_4^-]} = \frac{2f - 1}{2 - 2f}$$

Mass percent of acetic acid in the vinegar (extension only):

$$w = R\, c_t\, M \times 100$$

where R is the titrant-to-vinegar ratio in g/g, c_t is the amount of sodium hydroxide per gram of titrant solution in mol/g, and M is the molar mass of acetic acid in g/mol.

Materials

Chemicals

- ☐ Distilled white vinegar, labeled 5% acidity (Heinz or any brand with that label), about 10 mL
- ☐ Bar Keepers Friend Cleanser & Polish, the original powder in the can, about 2 g
- ☐ Lye, sold as crystal drain opener labeled 100% sodium hydroxide (Roebic Heavy Duty Crystal Drain Opener or similar, hardware or farm store), about 1 g. Do not use a drain product that lists aluminum or is sold as a gel or liquid.
- ☐ Distilled water, 1 gallon jug (grocery store)
- ☐ Ground turmeric, about 1 g (spice aisle)
- ☐ Rubbing alcohol, 70% or 91% isopropyl, about 10 mL (pharmacy)

Equipment

- ☐ Pocket scale reading to 0.01 g, capacity at least 100 g
- ☐ Universal pH test strips covering at least pH 3 to 10, with the finest color steps available (0.5 pH or finer). The accuracy of this lab is set by these strips.
- ☐ Plastic dropper bottle, 30 mL (1 oz) or larger, for the titrant
- ☐ Plastic bottle with cap, about 500 mL (a clean empty water bottle), for the sodium hydroxide stock
- ☐ Kitchen measuring cup, 1 cup or 250 mL
- ☐ Four clear plastic cups, about 250 mL (not foam; a 100 mL beaker may replace any of them)
- ☐ One small cup for the indicator
- ☐ Plastic transfer pipette or medicine dropper, for the vinegar
- ☐ Paper coffee filter and a funnel or a second cup to hold it
- ☐ Plastic spoon
- ☐ Measuring spoons, ¼ teaspoon and 1 teaspoon
- ☐ Safety goggles and nitrile or dishwashing gloves
- ☐ Paper towels
- ☐ Timer (phone)

Safety

> **Safety.** Wear safety goggles for the whole lab and gloves while handling the lye and its solutions. Lye (sodium hydroxide) causes severe chemical burns and permanent eye damage; it also heats water as it dissolves. Always add the lye to the water, never water to the lye, and keep it away from aluminum, which it attacks with release of hydrogen. The 0.1 M titrant is far more dilute but still irritates skin and eyes: rinse any contact with running water for 5 minutes, and rinse the eyes for 15 minutes and seek medical advice. Oxalic acid in the cleanser is toxic if swallowed and irritates skin and eyes; work away from food, do not reuse any lab cup for food, and wash hands at the end. Rubbing alcohol is flammable. Vinegar and turmeric are foods and are not hazards, though turmeric stains skin, plastic and clothing yellow.

Preparation

0.1 Measure **250 mL** of distilled water into the 500 mL bottle with the measuring cup. If a kitchen scale is available, weigh **249 g** of water instead. Set the bottle aside with the cap off.

0.2 Put a small dry cup on the pocket scale and tare it. Weigh **1.00 g** of lye crystals into the cup, working quickly because lye absorbs water from the air. Record the mass $\boldsymbol{m_{lye}}$ to 0.01 g.

0.3 Tip the lye into the water in the bottle, cap it, and swirl until no crystals remain. The bottle warms slightly. Label the bottle "0.1 M NaOH". Fill the dropper bottle from it and keep the rest capped.

0.4 Put **¼ teaspoon** of turmeric and **2 teaspoons** of rubbing alcohol in a small cup. Stir for 1 minute and let it stand for 5 minutes. The clear yellow liquid above the settled powder is the indicator. Draw indicator from the top of this cup when a step calls for it.

0.5 Put **2.0 g** of Bar Keepers Friend powder and about **60 mL** (¼ cup) of distilled water in a plastic cup. Stir for 5 minutes and let it stand for 5 minutes. The grey abrasive settles; the liquid stays slightly cloudy and may foam. Pour the liquid through a coffee filter into a clean cup. Collect at least 35 g of filtrate. This is the oxalic acid extract.

Measurements used in both parts

The titration. Swirl the sample cup so the indicator is evenly mixed; the solution is yellow. Make sure the dropper bottle holds at least 15 g of titrant, refilling from the stock bottle if it does not. Hold the dropper bottle upright so no drop hangs from the tip, place it on the scale, and record its mass $\boldsymbol{m_{before}}$ to 0.01 g. Add titrant to the sample a few drops at a time, swirling after each addition. When an orange or brown color appears where the drops land and takes more than a second to fade, slow to single drops. Stop at the first drop after which the whole solution stays orange-brown through 30 seconds of swirling. Hold the bottle upright, weigh it again, and record $\boldsymbol{m_{after}}$ to 0.01 g. The mass of titrant to the equivalence point is $m_{eq} = m_{before} - m_{after}$. Discard the run if a drop fell outside the cup or if a drop was left hanging from the tip at either weighing.

The pH reading. Stir the mixture with the plastic spoon for 10 seconds. Dip a fresh pH strip into the mixture for 1 second, shake off the excess, and compare it with the chart after the time stated on the package. Read to the strip's finest step, or to half a step if the color falls clearly between two blocks. Record the reading as $\mathbf{pH}_{\boldsymbol{read}}$. Do not put turmeric indicator in any mixture that will be read with a strip; the yellow dye changes the strip color.

Where the error comes from. Every mass in this lab is read to 0.01 g on samples of 1.5 g or more, and the endpoint is found to within one drop, about 0.05 g in 10 g. Together these fix the fraction f to better than 1 %, which moves the pH of a buffer by less than 0.01. The half-neutralized mixture is therefore made almost exactly. The only coarse measurement is the strip reading. A strip read to 0.5 pH gives pK_a to ±0.5 and K_a to within a factor of 3; a strip read to 0.25 gives K_a to within a factor of 1.8. The accuracy of this lab is the accuracy of the strips, so use the finest-graded strips available and read them carefully.

Part 1 — Acetic acid in vinegar

1.1 Put a clean plastic cup on the scale and tare it. Use the pipette to add **1.50 g** of vinegar, within 0.10 g. Record the mass $\boldsymbol{m_v}$ to 0.01 g.

1.2 Add about **20 mL** (4 teaspoons) of distilled water and **4 drops** of turmeric indicator to the cup. Swirl. The solution is pale yellow.

1.3 Make the titration. Record $\boldsymbol{m_{before}}$, $\boldsymbol{m_{after}}$ and $\boldsymbol{m_{eq}}$ in the table under Run 1.

1.4 Repeat steps 1.1 to 1.3 with a fresh vinegar sample as Run 2. Calculate $R = m_{eq} / m_v$ for each run. If the two values of R differ by more than 5% of their mean, do a third run and use the two closest values.

1.5 Put a clean plastic cup on the scale and tare it. Add **1.50 g** of vinegar, within 0.10 g, and record the mass $\boldsymbol{m_s}$ to 0.01 g. Add about **20 mL** (4 teaspoons) of distilled water. Do not add indicator.

1.6 Weigh the dropper bottle. Add titrant, re-weighing the bottle between additions, until the mass delivered is $\mathbf{0.25 \times R \times m_s}$, within 0.10 g. Record the mass added $\boldsymbol{m_t}$ to 0.01 g. Make the pH reading. Record $\mathbf{pH_{read}}$ in the table for the $f = 0.25$ point.

1.7 Continue adding titrant to the same cup until the total mass delivered since step 1.5 is $\mathbf{0.50 \times R \times m_s}$, within 0.10 g. Record $\boldsymbol{m_t}$ and make the pH reading for the $f = 0.50$ point.

1.8 Continue adding titrant until the total mass delivered is $\mathbf{0.75 \times R \times m_s}$, within 0.10 g. Record $\boldsymbol{m_t}$ and make the pH reading for the $f = 0.75$ point.

Photo. Photograph the two vinegar titration cups side by side at the end of Run 2, with the orange-brown endpoint color visible.

Part 2 — Oxalic acid in the cleanser extract

2.1 Put a clean plastic cup on the scale and tare it. Pour in **10 g** of the oxalic acid extract, within 1 g, and record the mass $\boldsymbol{m_v}$ to 0.01 g. Add **4 drops** of turmeric indicator and swirl. Foam from the detergent in the cleanser is normal; let it settle before reading colors.

2.2 Make the titration. There is no color change at the first equivalence point of this diprotic acid; the endpoint found here is the second equivalence point. Record $\boldsymbol{m_{before}}$, $\boldsymbol{m_{after}}$ and $\boldsymbol{m_{eq}}$ under Run 1.

2.3 Repeat steps 2.1 and 2.2 with a fresh 10 g portion of extract as Run 2. Calculate $R = m_{eq} / m_v$ for each run. If the two values differ by more than 5% of their mean, do a third run and

use the two closest values.

2.4 Put a clean plastic cup on the scale and tare it. Pour in **15 g** of extract, within 1 g, and record the mass $\boldsymbol{m_s}$ to 0.01 g. Do not add indicator.

2.5 Weigh the dropper bottle. Add titrant, re-weighing the bottle between additions, until the mass delivered is $\mathbf{0.625 \times R \times m_s}$, within 0.10 g. Record $\boldsymbol{m_t}$ and make the pH reading for the $f = 0.625$ point.

2.6 Continue adding titrant to the same cup until the total mass delivered since step 2.4 is $\mathbf{0.750 \times R \times m_s}$, within 0.10 g. Record $\boldsymbol{m_t}$ and make the pH reading for the $f = 0.750$ point.

2.7 Continue adding titrant until the total mass delivered is $\mathbf{0.875 \times R \times m_s}$, within 0.10 g. Record $\boldsymbol{m_t}$ and make the pH reading for the $f = 0.875$ point.

2.8 Pour all solutions down the drain with plenty of running water. Rinse the cups and the dropper bottle. The sodium hydroxide stock keeps for a month in the capped plastic bottle.

Photo. Photograph all six used pH strips laid on a white paper towel next to the package color chart, labeled with their f values.

From the fraction f and the strip reading at each point, calculate the pK_a of acetic acid and the second pK_a of oxalic acid. Compare the three estimates for each acid with one another and with the expected readings in the reference table, and compare the pK_a values with the accepted values. With the ratio R from Part 1 and the nominal titrant concentration, the mass percent of acetic acid in the vinegar can be calculated and compared with the label.

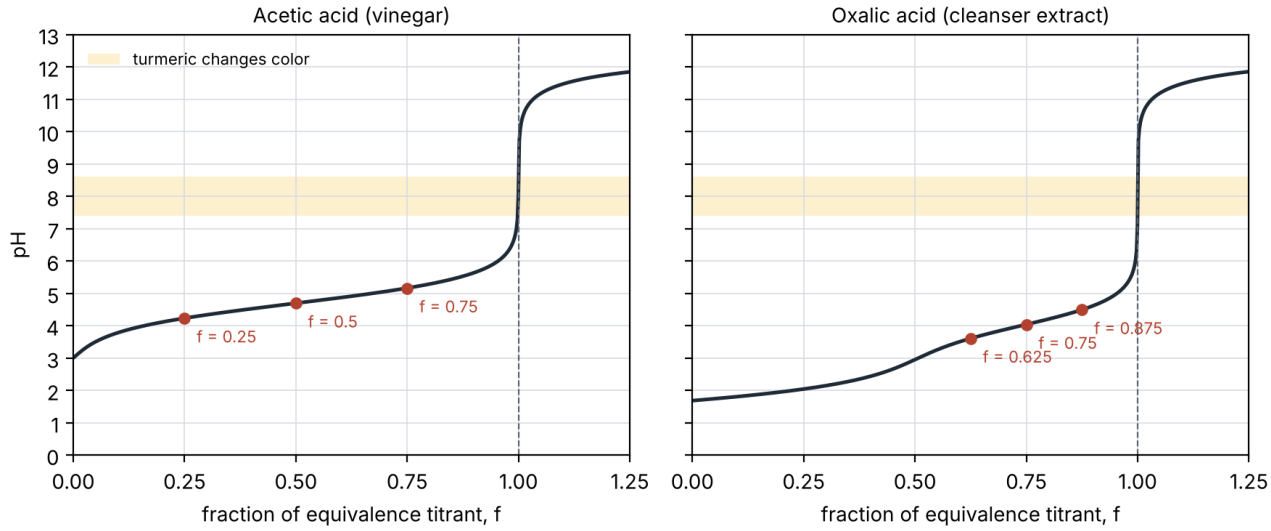
Reference material

Accepted values

Acid	Accepted pK_a at 25 °C, zero ionic strength	Source	Expected strip reading in this lab
Acetic acid	4.756	Harned and Ehlers, 1933	4.69 at f = 0.50
Oxalic acid, first dissociation	1.25	Martell and Smith, Critical Stability Constants, 1977	not measured
Oxalic acid, second dissociation	4.27	Martell and Smith, Critical Stability Constants, 1977	4.04 at f = 0.750

Why the expected reading is below the accepted pKa (optional). The accepted pK_a values refer to infinitely dilute solution. In the titration mixtures the dissolved ions, mainly sodium ions and the conjugate base, lower the activity of the charged species, and a pH meter or strip responds to activity. The mixtures here have an ionic strength near 0.02 mol/L for acetic acid and 0.04 mol/L for oxalic acid. The Davies equation then predicts a reading about 0.07 below the pK_a for acetic acid, whose conjugate base carries one charge, and about 0.23 below the second pK_a for oxalic acid, whose conjugate base carries two charges. The "expected strip reading" column already includes this shift, so a perfect strip in a perfect titration reads 4.69 and 4.04, not 4.76 and 4.27.

Model titration curves



Model pH during each titration at the sample sizes used in this lab, plotted against the fraction f of the equivalence titrant. Red points mark the three mixtures prepared for pH readings; the shaded band is where the turmeric indicator turns from yellow to orange-brown. Use it to check that each strip reading falls near the curve and to see why the oxalic acid curve shows no step at f = 0.5.

Sodium hydroxide concentration

The lye is labeled 100% sodium hydroxide, but crystals that have been opened absorb water and carbon dioxide from the air, so the titrant is nominally, not exactly, 0.100 mol/kg. The ratio R and every pK_a are independent of the titrant concentration. Only the extension (percent acetic acid in vinegar) depends on it, and the label value is uncertain to a few percent for that reason.

Data

Titrant: 1.00 g lye in 250 g solution, nominal $c_t = 1.00 \times 10^{-4}$ mol NaOH per g of titrant. Molar masses: acetic acid 60.05 g/mol, sodium hydroxide 40.00 g/mol. Vinegar label: 5.0% acidity.

Mass of lye weighed, m_{lye}: _____ g

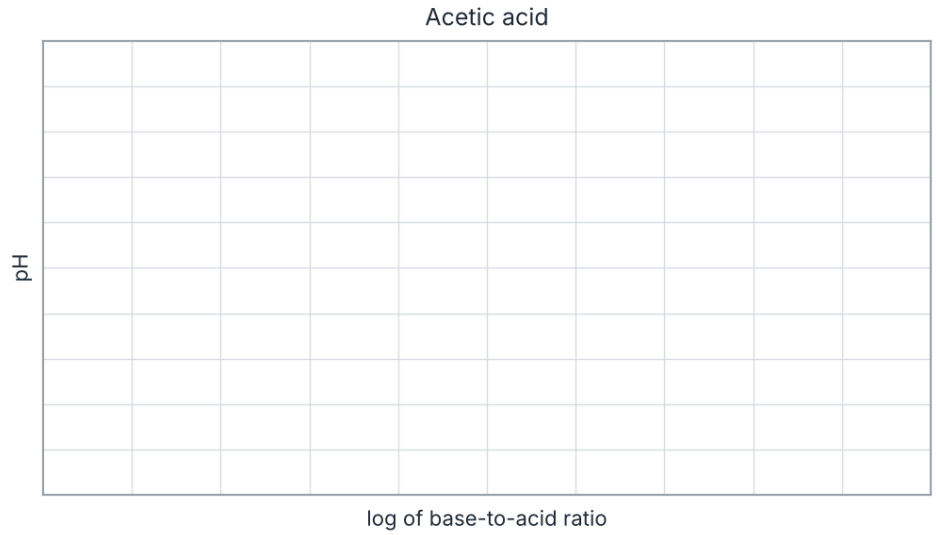
Equivalence titrations — acetic acid

Run	m_v (g)	m_{before} (g)	m_{after} (g)	m_{eq} (g)	R (g/g)
Run 1					
Run 2					
Run 3 (if needed)					

Mass of vinegar in the pH mixture, m_s: _____ g

pH readings — acetic acid

Point	Target f	m_t (g)	f	pH_{read}
Quarter	0.25			
Half	0.50			
Three quarters	0.75			



Graph. Henderson–Hasselbalch plot for acetic acid. The intercept is the pKa and the slope should be close to 1.

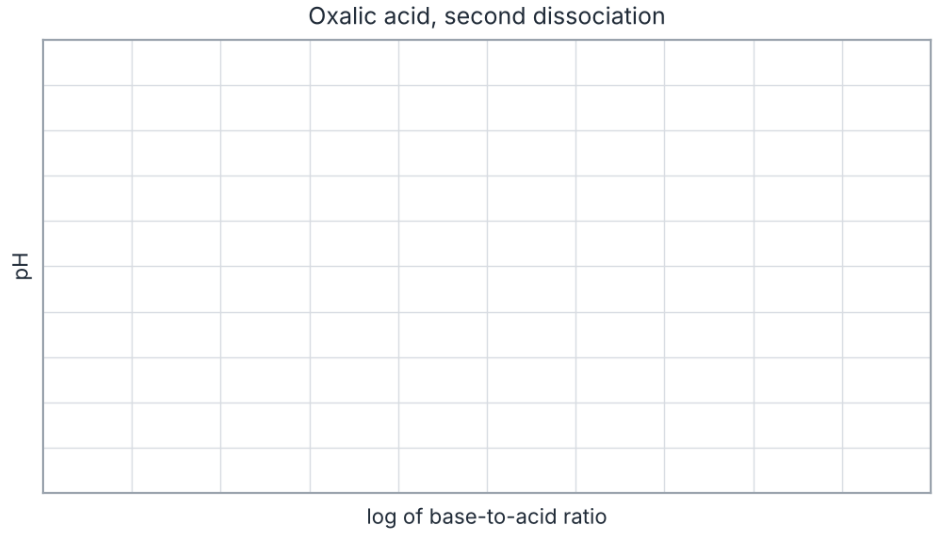
Equivalence titrations — oxalic acid extract

Run	m_v (g)	m_{before} (g)	m_{after} (g)	m_{eq} (g)	R (g/g)
Run 1					
Run 2					
Run 3 (if needed)					

Mass of extract in the pH mixture, m_s: _____ g

pH readings — oxalic acid extract

Point	Target f	m_t (g)	f	pH_{read}
Five eighths	0.625			
Three quarters	0.750			
Seven eighths	0.875			



Graph. Henderson–Hasselbalch plot for the second dissociation of oxalic acid. The intercept is the second pKa and the slope should be close to 1.

> **Style.** Write in the past tense and third person. Give every number a unit and a sensible number of significant figures. Refer to tables and figures by number.

Report

Purpose

State the goal of this experiment.

Hint. 30 to 80 words. State the two quantities that were measured and the property of the acids that each one describes.

Method

Describe what was done, in enough detail that another student could repeat it from the report alone.

Hint. 80 to 200 words. Describe how the equivalence point was found, how the partly neutralized mixtures were made, and how the pH was read.

Results

Summarize the measurements.

Hint. 60 to 150 words. State the ratio R for each acid, the three pH readings for each, and whether the readings rise with f as expected. Refer to the tables and figures below by number.

Include the table “Equivalence titrations — acetic acid” from the Data section, filled with the recorded values.

Include the table “pH readings — acetic acid” from the Data section, filled with the recorded values.

Include the graph “Acetic acid”, drawn from the recorded values.

Include the table “Equivalence titrations — oxalic acid extract” from the Data section, filled with the recorded values.

Include the table “pH readings — oxalic acid extract” from the Data section, filled with the recorded values.

Include the graph “Oxalic acid, second dissociation”, drawn from the recorded values.

Include the photo: Photograph all six used pH strips laid on a white paper towel next to the package color chart, labeled with their f values.

Calculations

Show the calculations that lead from the recorded masses and strip readings to the pK_a and K_a of each acid.

Hint. Show one worked example of each step for acetic acid, the fraction f from the masses, the base to acid ratio, the pKa from the Henderson Hasselbalch equation, and Ka from pKa. State the three pKa estimates for each acid and their mean. For oxalic acid use the ratio for the second dissociation.

Comparison with accepted values

Compare the results with the accepted values.

Hint. 80 to 200 words. Compare each mean pKa with the accepted value and with the expected strip reading in the reference table, and say which comparison is the fair one. State the percent difference. Compare the slope of each Henderson Hasselbalch plot with 1 and say what a slope far from 1 would mean.

Sources of error

Identify the sources of error in this experiment and the direction in which each one moves the result.

Hint. 100 to 250 words. The strip reading sets the accuracy of this lab, so say how far an error of one strip step moves the pKa and the Ka, and compare that with the effect of a 1 percent error in the titrant mass. Then consider the endpoint being overshot by a drop, the extract from the cleanser containing other dissolved material, and carbon dioxide absorbed by the titrant, and say for each which direction it pushes the pKa, not just that it exists.

Conclusion

State the conclusions of the experiment.

Hint. 50 to 150 words. Give the Ka of each acid with its uncertainty, say which acid is stronger and by what factor, and say whether the data support the claim that the pH at the half neutralized point equals the pKa.

Extension (optional)

Calculate the mass percent of acetic acid in the vinegar and compare it with the label.

Hint. 50 to 150 words. Use R for the vinegar and the nominal titrant concentration to find the mass percent of acetic acid, compare it with the 5.0 percent label, noting that the label is grams per 100 mL and vinegar is about 1 percent denser than water, and explain why the lye crystals make this value less certain than the pKa.